

Enhancing accuracy and interpretability in corporate credit rating classification with the transformer-LSTM model

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ABSTRACT

Corporate credit rating classification is essential for assessing a corporation's debt repayment ability. Previous research has demonstrated that neural network models exhibit high classification accuracy, particularly when incorporating time-series features. However, a significant challenge remains regarding their interpretability, often limited by nonlinear and intricate calculation processes. To address this trade-off between interpretability and the utilization of time-series features, we introduce a novel approach: the Transformer-Long Short-Term Memory (T-LSTM). Specifically, the attention matrix embedded within the T-LSTM architecture provides interpretability by revealing the temporal importance of features. Comparative experiments show that our T-LSTM model surpasses standard machine learning baselines in the long-history setting. Empirical results demonstrate that the proposed model, trained on a 10-year history of longitudinal financial ratios, yields an absolute accuracy improvement of approximately 1 percentage point when compared with a strong sequential baseline such as a Long Short-Term Memory (LSTM) model trained on the same 10-year history, and up to about 17 percentage points when compared with representative single-year "snapshot" baselines that use only the most recent year's ratios. Furthermore, the attention matrix successfully visualizes specific time points where information is most critical for rating classification. Consequently, the proposed model offers a highly accurate and interpretable solution for credit rating classification in the financial industry.

1. Introduction

Corporate credit risk assessment plays a pivotal role in investment decision-making and financial risk management. Credit analysts typically rely on a combination of quantitative and qualitative corporate attributes, derived principally from financial statements, to determine corporate credit ratings. In quantitative assessment, the Balance Sheet and the Income Statement are fundamental, providing the data necessary to calculate predictive indicators—such as the Current Ratio, Retained Earnings to Total Assets, and EBIT to Total Assets—that constitute foundational models like the Altman Z-Score.

To enhance the precision and efficiency of this evaluation process, various statistical models and machine learning methods have been employed. Early approaches utilized techniques such as logistic regression [5], Support Vector Machines (SVM) [2,6,7], and Random Forest [3]. More recently, advanced algorithms have established new benchmarks in the field. Gradient boosting methods, particularly LightGBM [24], are widely recognized as state-of-the-art for tabular financial data due to their speed and accuracy. In the realm of deep learning, novel

architectures such as TabNet [25], which combine the benefits of decision trees with neural network representation learning, and Temporal Convolutional Networks (TCN) [26], which offer a powerful alternative to recurrent architectures for sequence modeling, have been introduced. Consequently, these advanced methods serve as critical baselines for evaluating new proposals. Among these developments, neural networks remain a focal point due to their potential for capturing complex nonlinear patterns [1,4].

While prior research has traditionally focused on cross-sectional data using recently observed financial ratios, a company's repayment capability is significantly influenced by its historical trajectory, not merely its current condition [8,9]. Consequently, recent studies have shifted focus to longitudinal data, adopting Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models [10], which are architecturally optimized for handling time-series sequences. These studies have demonstrated that incorporating longitudinal financial features significantly enhances classification accuracy by capturing temporal dependencies [8,9].

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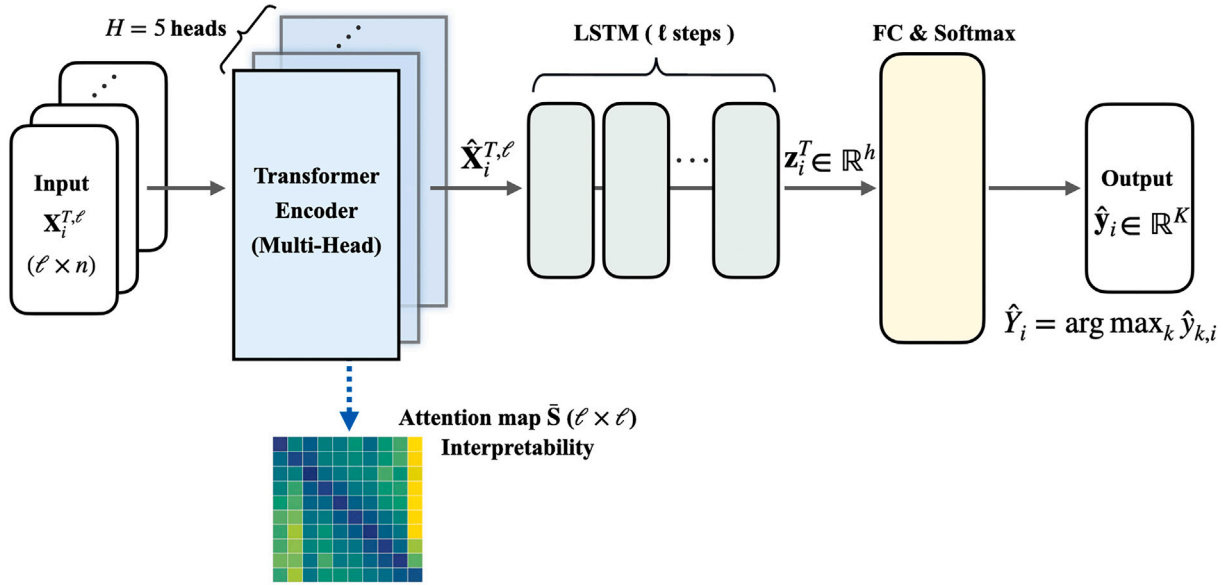


Fig. 1. Proposed T-LSTM architecture. A Transformer encoder with multi-head self-attention ($H = 5$) maps the input $\mathbf{X}_i^{T, \ell}$ to $\hat{\mathbf{X}}_i^{T, \ell}$ and provides an interpretable attention map $\tilde{\mathbf{S}} (\ell \times \ell)$. The LSTM then aggregates $\hat{\mathbf{X}}_i^{T, \ell}$ over ℓ steps to obtain $\mathbf{z}_i^T$, which is classified by an FC layer with softmax to produce $\hat{\mathbf{y}}_i$.

However, a critical challenge remains regarding the transparency of these advanced models. Although deep neural networks exhibit outstanding predictive performance, their calculation processes are inherently nonlinear and opaque. This “black box” nature poses significant hurdles in real-world financial applications, where the rationale behind credit decisions is essential for accountability and regulatory compliance. While post-hoc explanation methods such as SHAP values [11] or global surrogate models [12] have been introduced, they often provide approximations rather than intrinsic model transparency.

In this study, we propose a novel approach: the **Transformer-Long Short-Term Memory (T-LSTM)**. Our primary objective is to strike a balance between high classification accuracy and model interpretability. The Transformer, a state-of-the-art technique originally developed for natural language processing [13], plays a pivotal role in our approach by applying self-attention mechanisms to capture complex dependencies. Recent literature suggests that this architecture achieves state-of-the-art performance across various domains beyond NLP [14–16]. By leveraging the attention mechanism within the Transformer, our model intrinsically reveals which temporal points and financial features are critical for the decision-making process. This capability sharply distinguishes the T-LSTM from naive neural networks, as the attention matrix explicitly quantifies the importance of features along the time axis, thereby resolving the opacity inherent in traditional architectures. We integrate this with LSTM components to effectively capture long-term time-series dynamics, thereby ensuring robust and superior classification performance compared to conventional baselines.

The remainder of this paper is organized as follows. Section 2 presents the formulation of the corporate credit rating classification task and details the architecture of the proposed T-LSTM. Section 3 describes the dataset, including the longitudinal financial ratios and preprocessing steps, as well as the baseline models employed for comparison. Section 4 showcases the experimental results and provides an analysis of the temporal importance revealed by the Transformer’s attention matrix. Finally, Section 5 concludes the paper.

2. Proposed method

In this section, we introduce the formulation of corporate credit rating classification and T-LSTM, which is based on a Transformer and LSTM.

The credit rating classification model for financial time-series data aims to establish a function wherein the inputs consist of financial time-series data and the outputs correspond to credit ratings. Thus, given financial-statement time-series data $\mathbf{X}_i^{T, \ell} = \{\mathbf{x}_i^{T-\ell+1}, \dots, \mathbf{x}_i^T\}$ ($\mathbf{x}_i^{T-\ell+1} \in \mathbb{R}^n$) as input data, where i is an index of the corporate enterprise and n is the number of the financial indicators, we formulate the corporate credit rating classification as an estimation of the function $Y_i^T = F(\mathbf{X}_i^{T, \ell})$. $T \in \mathbb{N}$ and $\ell \in \{1, 2, \dots, T\}$ denote the time steps required to evaluate credit ratings and the length of the time-series data, respectively. $Y_i^T \in \{1, \dots, K\}$ denotes the predicted credit rating for input $\mathbf{X}_i^{T, \ell}$.

In this study, we propose the T-LSTM consisting of a Transformer and LSTM (Fig. 1) as the function F .

We assigned a Transformer to the first layer of our model to obtain interpretable parameters. In our model, each time step of the time-series data is processed by the Transformer to analyze the correlations among the inputs. Crucially, this architecture allows us to explain the rationale behind predictions by highlighting which parts of the input sequence are most influential.

First, we generate three matrices known as query, key, and value, from $\mathbf{X}_i^{T, \ell} = (\mathbf{x}_i^{T-\ell+1}, \dots, \mathbf{x}_i^T)^T \in \mathbb{R}^{\ell \times n}$ (where $\top$ indicates transpose), as follows:

$$\mathbf{Q} = \mathbf{X}_i^{T, \ell} \mathbf{W}_q \quad \mathbf{K} = \mathbf{X}_i^{T, \ell} \mathbf{W}_k \quad \mathbf{V} = \mathbf{X}_i^{T, \ell} \mathbf{W}_v, \quad (1)$$

where $\mathbf{W}_{(q,k,v)} \in \mathbb{R}^{n \times n}$ represent weight matrices. In summary, $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ can be conveniently generated by weighting the input features $\mathbf{X}_i^{T, \ell}$. These three matrices capture time-step features, which are then correlated to compute the attention weights as follows:

$$\mathbf{S} = \text{softmax} \left(\frac{\mathbf{QK}^\top}{\sqrt{n}} \right), \quad (2)$$

where the softmax function is expressed as

$$\text{softmax}(u_j) = \frac{\exp(u_j)}{\sum_{j'=1}^{\ell} \exp(u_{j'})}. \quad (3)$$

Here, the softmax function is applied along the rows of $\mathbf{QK}^\top$. In the multi-head self-attention setting, Eq. (2) is computed independently for each head m ($m = 1, \dots, H$) using the head-specific projections $\mathbf{Q}^{(m)}$

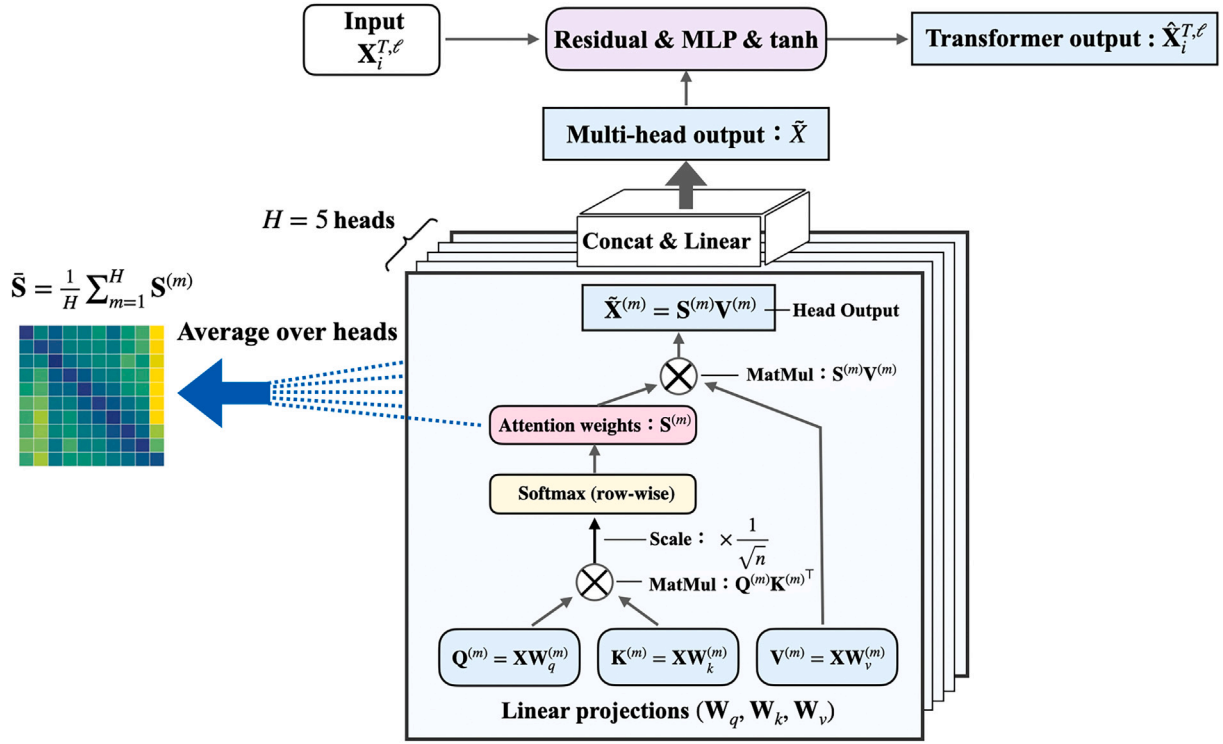


Fig. 2. Multi-head self-attention used in the Transformer encoder (single-head expanded). Each head m computes $S^{(m)} = \text{softmax}(Q^{(m)}K^{(m)\top}/\sqrt{n})$ (row-wise) and applies it to $V^{(m)}$; the head outputs are concatenated and linearly projected to obtain the multi-head output $\tilde{X}$. The final Transformer output $\hat{X}_i^{T,\ell}$ is then produced by the residual-MLP-tanh transform in Eq. (5). For visualization, we show the head-averaged attention map $\bar{S} = \frac{1}{H} \sum_{m=1}^H S^{(m)}$.

and $K^{(m)}$ to obtain $S^{(m)}$; for visualization, we report the head-averaged attention map $\bar{S} = \frac{1}{H} \sum_{m=1}^H S^{(m)}$.

The generated matrices $S^{(m)}$ serve as attention maps that quantify the relationships between different time steps. To visualize the multi-head self-attention computation, Fig. 2 presents a single-head expanded view, showing how $S^{(m)} = \text{softmax}(Q^{(m)}K^{(m)\top}/\sqrt{n})$ (row-wise) is used to weight $V^{(m)}$, and how the head outputs are concatenated and subsequently transformed by residual connections, an MLP, and $\tanh(\cdot)$ to obtain $\hat{X}_i^{T,\ell}$.

Crucially, the values in $\bar{S}$ reveal the *importance* of the features at each specific point in time. A higher value indicates that the information at that specific time step is more critical for the model's decision-making process. Therefore, the attention mechanism not only captures temporal dependencies but also provides interpretability by explicitly highlighting which historical financial data points are most relevant for the credit rating classification. We employ the attention weights S (or $S^{(m)}$ in the multi-head setting) to weight V in the following manner:

$$\tilde{X}_i^{T,\ell} = SV. \quad (4)$$

Then, residual connections [17] and nonlinear transforms are applied as follows:

$$\hat{X}_i^{T,\ell} = \tanh(X_i^{T,\ell} + \tilde{X}_i^{T,\ell} + \text{MLP}(X_i^{T,\ell} + \tilde{X}_i^{T,\ell})), \quad (5)$$

where the term “MLP” represents a nonlinear transformation performed by a multilayer perceptron (MLP). The MLP alters the number of units in the sequence $n \rightarrow 4n \rightarrow n$ through a single hidden layer of size $4n$ activated by ReLU. This transformation has been recommended for enhancing data feature representations [13]. As a result, we derive $\hat{X}_i^{T,\ell}$ as the output of the Transformer. $\hat{X}_i^{T,\ell}$ is interpreted as time-series data that incorporates the learned relationships among all time steps.

Multi-head attention is often employed when constructing a Transformer. We searched for the optimal multi-head size through cross-validation. Accordingly, we employ multi-head attention with five heads.

LSTM, originally introduced by Hochreiter and Schmidhuber [10], is a variant of Recurrent Neural Networks (RNNs) designed to capture long-term dependencies in sequential data while mitigating the vanishing gradient problem. In our proposed T-LSTM architecture, the LSTM layer serves as a temporal aggregator that processes the sequence of feature vectors extracted by the preceding Transformer layer.

Let $\hat{X}_i^{T,\ell} = \{\hat{x}_i^{T-\ell+1}, \dots, \hat{x}_i^T\}$ denote the sequence of output vectors from the Transformer, where $\hat{x}_i^t \in \mathbb{R}^n$. At each time step t , the LSTM unit updates its hidden state z^t and cell state s^t based on the current input $\hat{x}_i^t$ and the previous states (z^{t-1}, s^{t-1}). Unlike standard RNNs, the LSTM employs a gating mechanism to regulate the flow of information, effectively retaining relevant historical context over long sequences.

Through this recurrent process, the model generates a final hidden state vector $z_i^T \in \mathbb{R}^h$ at the last time step T . This vector z_i^T serves as a comprehensive representation of the corporate financial status, encapsulating the temporal dynamics identified across the entire time series.

Finally, to perform the corporate credit rating classification, we transform the vector z_i^T into class probabilities using a fully connected layer followed by a softmax function:

$$\hat{y}_i = \text{softmax}(Wz_i^T + \mathbf{b}), \quad (6)$$

where $W \in \mathbb{R}^{K \times h}$ and $\mathbf{b} \in \mathbb{R}^K$ represent the weight matrix and bias vector, respectively, and K denotes the number of target classes. The predicted credit rating $\hat{y}_i$ is determined by the class with the highest probability:

$$\hat{y}_i = \arg \max_{k \in \{1, \dots, K\}} (\hat{y}_{k,i}), \quad (7)$$

where $\hat{y}_{k,i}$ is the k -th element of $\hat{y}_i$.

Table 1
Mapping of R&I credit ratings to consolidated class labels and sample distribution.

Credit ratings	AAA ~ A+	A, A-	BBB+ ~ D
Class labels	0	1	2
Sample size	1735	2429	1669

3. Experimental setup

In this section, we detail the datasets, the preprocessing pipeline, the data partitioning strategy, the sequence generation methodology, and the configurations of the models employed in our experiments. We conducted two distinct experiments to evaluate the models under different temporal conditions: a primary experiment predicting 2019 ratings, and a supplementary experiment predicting 2018 ratings as an earlier test year. To ensure reproducibility, fixed random seeds were utilized: 209 for the 2019 experiment and 229 for the 2018 experiment.

3.1. Dataset

We utilized a structured dataset derived from the AstraManager database, provided by Quick Inc. This proprietary dataset integrates publicly disclosed corporate financial statements of Japanese listed firms with credit ratings meticulously assigned by Rating and Investment Information, Inc. (R&I). The data are structured as an annual firm-year panel, with each observation corresponding to a specific firm in a given fiscal year.¹

For the purpose of model development and rigorous evaluation, the comprehensive dataset was partitioned as follows:

- **Test set (Evaluation data):** The test set was constructed using credit rating samples from the fiscal years **2018 and 2019**. This period was reserved for the **final, out-of-sample evaluation** to assess the generalization capability of the developed models.
- **Training set (Development data):** The training data comprised credit rating samples spanning the years **1999 to 2017** and **1999 to 2018**. Specifically, the **1999–2017 period** served as the primary **training and validation set** for model calibration and hyperparameter tuning, while the **1999–2018 period** was utilized in specific cross-validation or robustness checks, depending on the model architecture. These sets were used to form the basis for our **model parameter estimation** and **final performance assessment**, respectively.

Based on these partitions, we considered two distinct evaluation scenarios. The first scenario, which provides an earlier test-year benchmark, utilizes the 1999–2017 training data to predict the 2018 test samples. The second scenario, regarded as the main evaluation, extends the training period to 1999–2018 to predict the 2019 test samples. This temporal hold-out design ensures that all models are evaluated on out-of-sample data that are temporally subsequent to the training data.

The original R&I credit ratings span multiple grades from AAA to D. Given the significant imbalance inherent in these grades, we consolidated the ratings into three ordinal classes, as delineated in Table 1. Our classification task focuses on predicting these consolidated class labels.

Consistent with prior research in corporate credit rating prediction, we employed 76 financial indicators as explanatory variables. These

indicators encompass key financial aspects such as earnings, leverage, profitability, cash flow, and turnover ratios. A comprehensive list and descriptive statistics of these indicators are provided in Table 2, and their definitions are detailed in Appendix A.

3.2. Preprocessing

All explanatory variables were converted to numeric format, with non-numeric values coerced to missing values (NaN). Observations were sorted chronologically by firm identifier and fiscal year (denoted as “time_index”). For each firm i , the feature sequence is denoted by $\{x_{i,t}\}_{t=1}^T$, where $x_{i,t} \in \mathbb{R}^{76}$.

Missing value imputation: Missing values were generally addressed using a two-stage procedure: (1) Forward filling longitudinally within each firm; (2) Imputation of remaining missing values using the median of the respective indicator, computed exclusively using the training data. In the 2018 experiment, benchmark models (TCN, TabNet, MLP, SVM, RF, LGBM) employed a stricter imputation strategy designed to prevent potential lookahead bias during forward filling, ensuring test data features were temporarily masked during the initial application on the training data.

Feature scaling (Cross-sectional standardization): Given the diverse scales of financial indicators and the need to adjust for market-wide temporal trends, we applied cross-sectional standardization. This method standardizes features at each specific time index t :

$$x'_{i,t,j} = \frac{x_{i,t,j} - \mu_{t,j}}{\sigma_{t,j}}, \quad (8)$$

where $x_{i,t,j}$ denotes the value of indicator j for firm i at time t .

To rigorously prevent look-ahead bias, the statistics $\mu_{t,j}$ and $\sigma_{t,j}$ were derived **exclusively from the training set**. For time steps within the training period, the statistics were calculated using the available training samples at that specific time t . For the test period (e.g., fiscal year 2018 or 2019), where no training samples are available by design, the statistics from the most recent training time step were applied. These training-derived statistics were then used to standardize the test samples, ensuring that no distributional information from the test period was leaked into the model inputs.

3.3. Data partitioning and sequence generation

Time-based validation split: To facilitate hyperparameter tuning and early stopping while maintaining the temporal order, we employed a time-based split for validation within the development data. We identified a split time point such that approximately 80% of the unique time indices were allocated to the training subset, and the subsequent (most recent) 20% were allocated to the validation subset.

Sequence generation: For sequence generation, let $\{x'_{i,t}\}_{t=1}^T$ represent the standardized feature sequence for firm i . To investigate the influence of temporal context, we generated sequences of five distinct lengths:

$$\ell \in \{1, 3, 5, 7, 10\}.$$

For each firm i and time index $t \geq \ell$, we constructed the input matrix (tensor)

$$X_i^{(t,\ell)} = (x'_{i,t-\ell+1}, \dots, x'_{i,t}) \in \mathbb{R}^{\ell \times 76}.$$

A sequence $X_i^{(t,\ell)}$ was assigned to the training, validation, or test set based on the affiliation of the corresponding observation (i, t) . The class label corresponds to the consolidated rating observed at the final time step t . In the 2018 experiment, for benchmark models, a specific leakage prevention mechanism was implemented to exclude training sequences that contained data points overlapping with the test period.

¹ The dataset used in this study is proprietary and was provided by Quick Inc. under a commercial license via the AstraManager service. Therefore, the underlying firm-level data tables cannot be made publicly available. Researchers interested in replication should obtain access directly from Quick Inc. through a valid AstraManager subscription. Data access is strictly contingent upon the express permission of Quick Inc.

Table 2

Descriptive statistics of the 76 financial indicators used as explanatory variables. This table presents the variable symbol, unit, range (minimum and maximum values), and mean value for each indicator. The detailed definitions and calculation formulas for each indicator are provided in [Appendix A](#).

Indicator	Symbol	Unit	Range (Min – Max)	Mean $\pm$ S.D.
Net sales	X_1	M JPY	6,896.00 – 30,225,681.00	939,853.09 $\pm$ 2,000,455.13
Ordinary profit	X_2	M JPY	– 560,381.00 – 2,983,381.00	59,096.64 $\pm$ 153,269.21
Total assets	X_3	M JPY	19,848.00 – 51,936,949.00	1,221,681.21 $\pm$ 2,676,094.66
Owners' equity	X_4	M JPY	– 129,483.00 – 19,348,152.00	414,682.76 $\pm$ 927,741.76
Cash flows from operating activities	X_5	M JPY	– 343,019.00 – 8,423,259.00	86,404.23 $\pm$ 252,610.47
Net working capital	X_6	M JPY	– 957,652.00 – 4,786,009.00	128,308.35 $\pm$ 328,071.22
Current ratio	X_7	Ratio	0.22 – 9.14	1.67 $\pm$ 0.93
Asset ratio	X_8	Ratio	0.02 – 6.04	1.01 $\pm$ 0.64
Current liabilities to cash flows from operating activities ratio	X_9	Ratio	– 0.81 – 3.32	0.29 $\pm$ 0.24
Fixed liabilities to cash flows from operating activities ratio	X_{10}	Ratio	– 4.76 – 57.81	0.61 $\pm$ 1.54
Interest-bearing liabilities to cash flows from operating activities ratio	X_{11}	Ratio	– 48.19 – 6,451.50	4.62 $\pm$ 86.33
Debt-equity ratio	X_{12}	Ratio	– 37.21 – 347.47	2.12 $\pm$ 4.84
Equity ratio	X_{13}	Ratio	– 0.16 – 0.89	0.42 $\pm$ 0.18
Fixed assets to equity ratio	X_{14}	Ratio	– 14.05 – 138.48	1.74 $\pm$ 2.33
Fixed assets to long-term capital ratio	X_{15}	Ratio	0.05 – 7.74	0.82 $\pm$ 0.28
Interest coverage ratio	X_{16}	Ratio	– 30,690.83 – 45,573.00	144.86 $\pm$ 1,267.56
Current liabilities to sales ratio	X_{17}	Ratio	0.01 – 15.35	3.01 $\pm$ 1.30
Fixed liabilities to sales ratio	X_{18}	Ratio	0.04 – 219.16	6.90 $\pm$ 12.29
Interest-bearing liabilities to sales ratio	X_{19}	Ratio	0.09 – 36,394.25	40.25 $\pm$ 650.10
Change ratio to sales ratio	X_{20}	Ratio	– 0.91 – 10.48	0.04 $\pm$ 0.25
Change ratio of total profit	X_{21}	Ratio	– 9.70 – 25.30	0.06 $\pm$ 0.55
Change ratio of operating profit	X_{22}	Ratio	– 455.61 – 92.94	0.09 $\pm$ 6.21
Change ratio of ordinary profit	X_{23}	Ratio	– 949.38 – 1,024.61	0.02 $\pm$ 17.02
Change ratio of total assets	X_{24}	Ratio	– 0.36 – 2.98	0.04 $\pm$ 0.12
Change ratio of owners' equity	X_{25}	Ratio	– 35,606.44 – 195,406.00	553.60 $\pm$ 5,115.28
Gross profit margin	X_{26}	Ratio	– 0.09 – 1.00	0.27 $\pm$ 0.15
Operating profit to sales ratio	X_{27}	Ratio	– 1.22 – 0.62	0.07 $\pm$ 0.06
Ordinary profit to sales ratio	X_{28}	Ratio	– 1.21 – 0.65	0.07 $\pm$ 0.07
Pretax income to sales ratio	X_{29}	Ratio	– 1.33 – 0.76	0.06 $\pm$ 0.07
Administrative expense ratio	X_{30}	Ratio	0.00 – 2.10	0.20 $\pm$ 0.14
Ordinary profit to total capital ratio	X_{31}	Ratio	– 0.26 – 0.38	0.05 $\pm$ 0.04
Operating profit to total capital ratio	X_{32}	Ratio	– 0.24 – 0.40	0.05 $\pm$ 0.04
Business profit to total capital ratio	X_{33}	Ratio	– 0.23 – 0.40	0.06 $\pm$ 0.04
Ordinary profit to owners' equity ratio	X_{34}	Ratio	– 3.68 – 6.94	0.13 $\pm$ 0.16
Operating profit to owners' equity ratio	X_{35}	Ratio	– 3.23 – 12.27	0.13 $\pm$ 0.19
Cash flows from operating activities to sales ratio	X_{36}	Ratio	– 1.14 – 2.08	0.09 $\pm$ 0.09
Total assets turnover	X_{37}	Ratio	0.05 – 3.21	0.86 $\pm$ 0.38
Equity turnover	X_{38}	Ratio	– 57.18 – 215.45	2.58 $\pm$ 3.79
Trade receivables turnover	X_{39}	Ratio	0.14 – 54,552.00	16.16 $\pm$ 632.91
Inventories turnover	X_{40}	Ratio	0.17 – 43,865.29	39.23 $\pm$ 536.46
Current assets turnover	X_{41}	Ratio	0.00 – 12.11	2.14 $\pm$ 1.23
Fixed assets turnover	X_{42}	Ratio	0.01 – 18.58	1.90 $\pm$ 1.48
Change rate of liabilities	X_{43}	Ratio	– 0.58 – 4.13	0.03 $\pm$ 0.19
Change rate of interest-bearing liabilities	X_{44}	Ratio	– 1.00 – 3,849.34	0.71 $\pm$ 44.90
Change rate of cash and deposits	X_{45}	Ratio	– 0.89 – 11.12	0.11 $\pm$ 0.45
Change rate of fixed assets	X_{46}	Ratio	– 0.92 – 2.57	0.05 $\pm$ 0.16
Cash to deposits ratio	X_{47}	Ratio	0.22 – 489.65	42.31 $\pm$ 44.88
Discounted interest expense to sales ratio	X_{48}	Ratio	0.00 – 21.08	0.70 $\pm$ 1.18
Total profit to discounted interest expense ratio	X_{49}	Ratio	– 21,511.54 – 2,968,817.09	92,170.35 $\pm$ 182,226.31
Tangible fixed assets to total assets ratio	X_{50}	Ratio	0.27 – 93.20	35.88 $\pm$ 19.04
Operating expenses to sales ratio	X_{51}	Ratio	2.01 – 95.84	46.11 $\pm$ 17.98
Other income to sales ratio	X_{52}	Ratio	0.35 – 96.94	53.52 $\pm$ 18.05
Other expenses to sales ratio	X_{53}	Ratio	0.00 – 53.60	10.50 $\pm$ 7.32
Interest burden to sales ratio	X_{54}	Ratio	– 29.00 – 37.93	4.15 $\pm$ 4.45
Ratio of current liabilities to cash flows from operating activities	X_{55}	Ratio	9.90 – 17.77	13.21 $\pm$ 1.23
Ratio of current liabilities to owners' equity	X_{56}	Ratio	– 815.81 – 10,038.64	121.54 $\pm$ 182.87
Total capital growth ratio	X_{57}	Ratio	– 128.50 – 5,305.92	21.77 $\pm$ 64.94
Pretax net earnings for the period to total capital ratio	X_{58}	Ratio	– 36.48 – 349.17	3.19 $\pm$ 4.86
Tangible fixed assets growth ratio	X_{59}	Ratio	0.08 – 59.27	3.64 $\pm$ 4.06
Liabilities turnover	X_{60}	Ratio	– 168.48 – 1,343.63	0.06 $\pm$ 17.12
Current liabilities turnover period	X_{61}	Years	– 1,308,055.56 – 596,412.50	364.85 $\pm$ 18,999.22
Short-term borrowings turnover period	X_{62}	Years	– 480,344.44 – 837,112.50	707.53 $\pm$ 12,310.64
Capital turnover period	X_{63}	Years	– 2,888.52 – 983.62	5.11 $\pm$ 41.97
Fixed liabilities turnover period	X_{64}	Years	– 625.98 – 4,587.75	9.52 $\pm$ 56.82
Days cash and deposits on hand	X_{65}	Days	0.42 – 221.62	62.63 $\pm$ 42.28
Cash interval	X_{66}	Days	0.01 – 48.75	1.01 $\pm$ 1.10
Current assets to total assets ratio	X_{67}	Ratio	0.01 – 21.97	1.27 $\pm$ 1.49
Fixed assets to total assets ratio	X_{68}	Ratio	0.00 – 26.14	0.71 $\pm$ 1.45
Inventories to total assets ratio	X_{69}	Ratio	– 97.66 – 279.88	3.16 $\pm$ 14.54
Pretax net earnings for the period to total assets ratio	X_{70}	Ratio	0.01 – 19.56	0.92 $\pm$ 1.20
Logarithm of total assets	X_{71}	log(M JPY)	0.01 – 9.55	0.44 $\pm$ 0.50
Tangible fixed assets ratio	X_{72}	Ratio	0.00 – 2.59	0.08 $\pm$ 0.14
Owners' equity to legal capital	X_{73}	Ratio	– 0.01 – 6.33	0.46 $\pm$ 0.44
Financial leverage ratio	X_{74}	Ratio	0.00 – 10.36	0.45 $\pm$ 0.69
Tangible fixed assets turnover	X_{75}	Ratio	0.00 – 2.48	0.14 $\pm$ 0.14
Net income growth ratio	X_{76}	Ratio	0.00 – 140.31	0.95 $\pm$ 1.91

Table 3

Performance comparison across sequence lengths $\ell \in \{1, 3, 5, 7, 10\}$ on the 2018 test data. Bold indicates the results of the proposed T-LSTM model.

Model	$\ell = 1$		$\ell = 3$		$\ell = 5$		$\ell = 7$		$\ell = 10$	
	AC	F1	AC	F1	AC	F1	AC	F1	AC	F1
T-LSTM	0.7862	0.7853	0.8007	0.8025	0.7790	0.7778	0.8623	0.8650	0.8938	0.8931
LSTM	0.7826	0.7832	0.7681	0.7723	0.8370	0.8389	0.8514	0.8539	0.8791	0.8802
TCN	0.6993	0.7034	0.7391	0.7416	0.7826	0.7867	0.8188	0.8206	0.8462	0.8487
TabNet	0.6667	0.6629	0.6848	0.6848	0.6848	0.6722	0.6884	0.6828	0.6996	0.6717
MLP	0.7355	0.7326	0.7609	0.7606	0.7935	0.7937	0.8152	0.8175	0.8278	0.8278
SVM	0.6848	0.6745	0.7138	0.7017	0.6232	0.5640	0.5978	0.5004	0.5311	0.3668
RF	0.7210	0.7223	0.8514	0.8556	0.8514	0.8532	0.8732	0.8769	0.8645	0.8667
LGBM	0.7029	0.7068	0.8297	0.8316	0.8297	0.8319	0.8514	0.8540	0.8681	0.8700

3.4. Model configurations and implementation

We evaluated the classification performance of T-LSTM against a comprehensive suite of benchmark models.

The specialized deep learning architectures (T-LSTM, LSTM, TCN, and TabNet) were implemented using the PyTorch framework and trained using the Adam optimizer [18]. Early stopping based on the validation loss (with patience ranging from 5 to 15 epochs) was employed. Baseline models (MLP, SVM, RF, and LGBM) were implemented using the scikit-learn library.

3.4.1. Sequential models

These models (T-LSTM, LSTM, and TCN) directly process the sequential input tensors $X_i^{(t,\ell)}$.

T-LSTM and LSTM: The T-LSTM architecture integrates a simple self-attention block and an LSTM block, as detailed in Section 2. We configured the number of LSTM hidden units to $h = 150$. The learning rate was set to 10^{-3} . Training utilized a batch size of 128 for a maximum of 50 epochs, with an early stopping patience of 5. A standard LSTM model with the same configuration was used for comparison.

TCN (Temporal convolutional network): The architecture comprises 3 layers of temporal blocks with channel dimensions of [64, 64, 64]. We used a kernel size of 3 and dilation factors of [1, 2, 4]. A dropout rate of 0.2 was applied. The model was trained with a batch size of 128, a learning rate of 10^{-3} , for a maximum of 50 epochs, with a patience of 5.

3.4.2. Non-sequential models

For these models (TabNet, MLP, SVM, RF, and LGBM), each input tensor $X_i^{(t,\ell)} \in \mathbb{R}^{\ell \times 76}$ was flattened into a vector of dimension 76ℓ .

TabNet: The hyperparameters were configured as follows: $N_d = N_a = 32$, $N_{steps} = 5$, $\gamma = 1.3$, and $\lambda_{sparse} = 10^{-4}$. We employed a learning rate of 2×10^{-2} with a StepLR scheduler (step size 10, gamma 0.9). Training utilized a batch size of 1024 (virtual batch size 128) for up to 100 epochs, with a patience of 15.

MLP (Multilayer perceptron): We implemented a standard MLP classifier with two hidden layers ([256, 128] units) and ReLU activation. It was trained using the Adam optimizer with a batch size of 128 for a maximum of 100 iterations.

SVM (Support vector machine): We employed SVM using the One-vs-Rest (OvR) strategy for multi-class classification [19,20] with the Radial Basis Function (RBF) kernel:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2). \quad (9)$$

We set the kernel coefficient $\gamma = 0.03$ and the regularization parameter $C = 32$.

RF (Random forest) and LGBM (LightGBM): We used RF with 1000 estimators and a maximum depth of 2048. Additionally, we employed LGBM [21], configured with 300 estimators, a maximum depth of 10, 130 leaves, and a learning rate of 0.05.

3.5. Evaluation metrics

We evaluated the classification performance of all models using Accuracy (AC) and Macro-Averaged F1-score ($F1_{macro}$).

Accuracy measures the overall proportion of correctly classified instances:

$$AC = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

Given potential class imbalances [22], we also utilize the Macro-F1 score ($F1_{macro}$) [23], which weights all classes equally. It is computed as the harmonic mean of the macro-averaged precision (PRE_{macro}) and recall (REC_{macro}):

$$F1_{macro} = \frac{2 \cdot PRE_{macro} \cdot REC_{macro}}{PRE_{macro} + REC_{macro}}, \quad (10)$$

where:

$$PRE_{macro} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FP_k}, \quad (11)$$

$$REC_{macro} = \frac{1}{K} \sum_{k=1}^K \frac{TP_k}{TP_k + FN_k}. \quad (12)$$

Here, k denotes the class index, K is the total number of classes ($K = 3$), and TP_k , FP_k , and FN_k represent the True Positives, False Positives, and False Negatives for class k , respectively.

4. Experiments and discussion

In this section, we evaluate the proposed T-LSTM model in terms of predictive performance and interpretability. We first compare its classification accuracy with that of several benchmark models, including state-of-the-art methods such as LightGBM, TabNet, and TCN, under different sequence lengths, and then analyze the attention mechanism to clarify which parts of the financial histories contribute most to the credit rating decisions.

4.1. Classification performance of proposed t-LSTM model

We evaluated the models across varying sequence lengths $\ell \in \{1, 3, 5, 7, 10\}$. The results for the 2018 prediction task (trained on 1999–2017 data) and the 2019 prediction task (trained on 1999–2018 data) are presented separately, followed by a detailed discussion.

4.1.1. Results of the 2018 prediction experiment

Table 3 summarizes the Accuracy (AC) and $F1_{macro}$ (F1) scores for the 2018 test set. Fig. 3 visualizes these results for selected sequence lengths ($\ell = 1, 5, 10$).

For the 2018 experiment, most models benefit from incorporating longer time-series information. The accuracy of T-LSTM increases significantly from 0.7862 at $\ell = 1$ to 0.8938 at $\ell = 10$. This confirms that



Fig. 3. Comparison of Accuracy (left column) and $F1_{macro}$ (right column) scores for the 2018 experiment ($\ell \in \{1, 5, 10\}$).

Table 4

Performance comparison across sequence lengths $\ell \in \{1, 3, 5, 7, 10\}$ (2019 Primary Experiment). Bold indicates the results of the proposed T-LSTM model.

Model	$\ell = 1$		$\ell = 3$		$\ell = 5$		$\ell = 7$		$\ell = 10$	
	AC	F1	AC	F1	AC	F1	AC	F1	AC	F1
T-LSTM	0.8231	0.8230	0.8454	0.8458	0.8454	0.8433	0.8996	0.9011	0.8967	0.8974
LSTM	0.8393	0.8388	0.8265	0.8249	0.8371	0.8374	0.8867	0.8881	0.8830	0.8847
TCN	0.7751	0.7742	0.7886	0.7871	0.7450	0.7461	0.7215	0.7244	0.6488	0.6327
TabNet	0.7321	0.7249	0.7047	0.7054	0.6780	0.6719	0.6970	0.6838	0.6151	0.5990
MLP	0.7975	0.7951	0.7868	0.7869	0.8147	0.8149	0.8040	0.8052	0.8003	0.7995
SVM	0.7670	0.7651	0.7685	0.7611	0.6952	0.6690	0.5990	0.5114	0.5291	0.3661
RF	0.8112	0.8080	0.8297	0.8270	0.8454	0.8433	0.8478	0.8454	0.8459	0.8445
LGBM	0.8280	0.8260	0.8397	0.8370	0.8479	0.8478	0.8601	0.8587	0.8591	0.8567

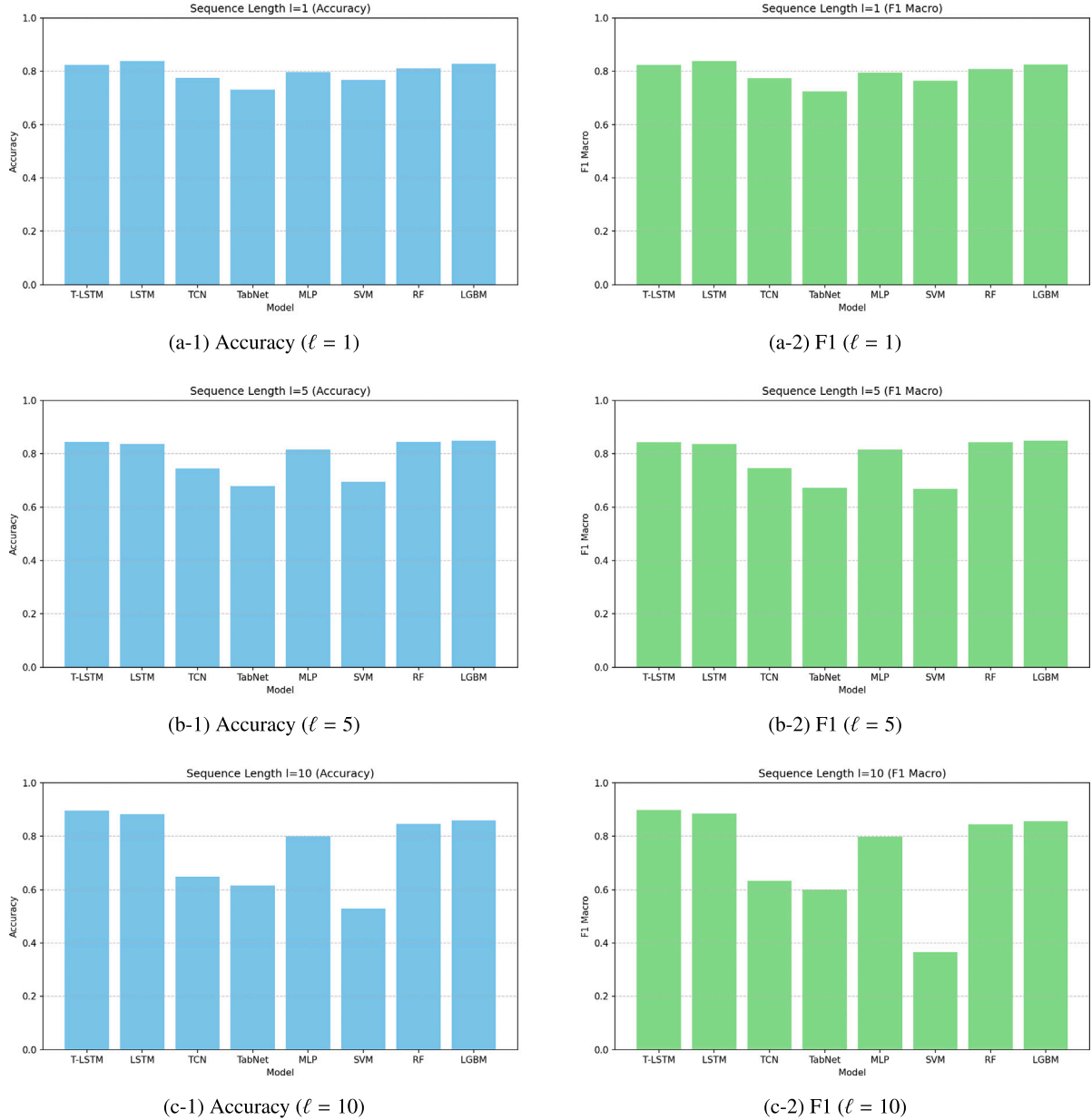


Fig. 4. Comparison of Accuracy (left column) and $F1_{macro}$ (right column) scores for the 2019 experiment ($\ell \in \{1, 5, 10\}$).

historical financial trajectories carry additional information beyond the most recent financial statement.

The relative ranking among models depends on the sequence length. While tree-based ensembles (RF and LGBM) are strong baselines for intermediate lengths, T-LSTM clearly achieves the best performance when the full 10-year history is used ($\ell = 10$). At this length, T-LSTM outperforms not only standard LSTM and TCN but also the strongest tabular baselines (RF and LGBM), illustrating its ability to make more effective use of long-term dependencies.

4.1.2. Results of the 2019 prediction experiment

Table 4 and Fig. 4 present the results for the 2019 test set.

The 2019 experiment shows a similar pattern. T-LSTM improves from 0.8231 at $\ell = 1$ to 0.8967 at $\ell = 10$, achieving peak-level performance for long sequences. Although the maximum accuracy is observed at $\ell = 7$ (0.8996), the performance at $\ell = 10$ remains comparable (0.8967). Moreover, when we consider both test years, the decade-long setting

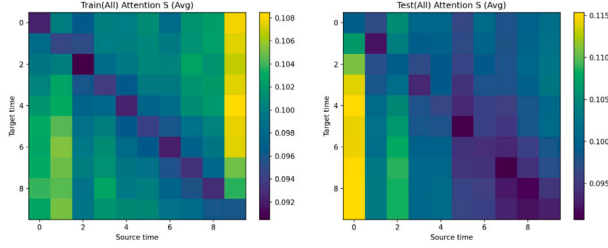
($\ell = 10$) provides the most consistently strong performance (e.g., the average accuracy across the two experiments is 0.8953 at $\ell = 10$, compared to 0.8810 at $\ell = 7$).

Overall, the two experiments indicate that the proposed T-LSTM model provides stable and high performance across different test years. When a long financial history is available (in particular, $\ell = 10$), T-LSTM achieves accuracies close to 0.90 and consistently outperforms advanced baselines.

4.1.3. Discussion on model performance and sequence length

The analysis across varying sequence lengths ℓ reveals several insights into the nature of corporate credit rating classification and the behavior of different model architectures.

Impact of sequence length and long-term dependencies. For most models, increasing ℓ from 1 to 10 improves performance, indicating that credit ratings are influenced by long-term financial trajectories rather



(a) Training data

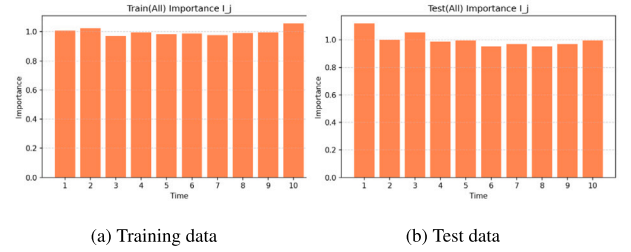
(b) Test data

Fig. 5. Heat map of the averaged attention matrix $\bar{S}$ for the training and test data (2019 Experiment).

than by a single year's ratios. This effect is particularly pronounced for T-LSTM, confirming that long-term dependencies are crucial for assessing corporate creditworthiness.

The substantial gains achieved by our approach underscore the value of integrating long-term longitudinal data. To quantify this effect in a way consistent with the traditional “current-snapshot” setting, we compare the proposed T-LSTM using a 10-year history ($\ell = 10$) with benchmark models using only the most recent year ($\ell = 1$). For example, in the 2018 experiment, T-LSTM at $\ell = 10$ (AC: 0.8938) outperforms RF at $\ell = 1$ (AC: 0.7210) by approximately 17 percentage points. In the 2019 experiment, T-LSTM at $\ell = 10$ (AC: 0.8967) outperforms TabNet at $\ell = 1$ (AC: 0.7321) by 16.46 percentage points. At the same time, even under the same long-history setting ($\ell = 10$), T-LSTM maintains a measurable advantage over strong baselines (e.g., 0.8938 vs. 0.8791 for LSTM in 2018), corresponding to an improvement on the order of 1 percentage point. Overall, these comparisons clarify that the reported gains depend on the evaluation setting: the improvement is on the order of 1 percentage point when compared with a strong sequential baseline under the same $\ell = 10$ setting (e.g., LSTM), whereas it reaches roughly 16–17 percentage points when compared with representative single-year “snapshot” baselines under $\ell = 1$.

Advantages of T-LSTM over other architectures. The comparison with LSTM clarifies the role of the Transformer layer. While a standard LSTM already improves as ℓ increases, T-LSTM yields higher accuracies, particularly when ℓ is extended to 10. This suggests that the self-attention mechanism helps the model focus on the most informative years, making the subsequent LSTM layer more effective at aggregating long-term patterns. TCN, on the other hand, fails to keep up with LSTM-based models for longer sequences, despite being widely recognized as a powerful alternative to recurrent architectures for sequence modeling. This suggests that the convolutional architecture may have difficulty



(a) Training data

(b) Test data

Fig. 6. Aggregated time step importance I_j ($1 \leq j \leq 10$) for the training and test data (2019 Experiment).

capturing the irregular long-horizon dynamics that are better handled by the gated recurrence in LSTM, enhanced by attention.

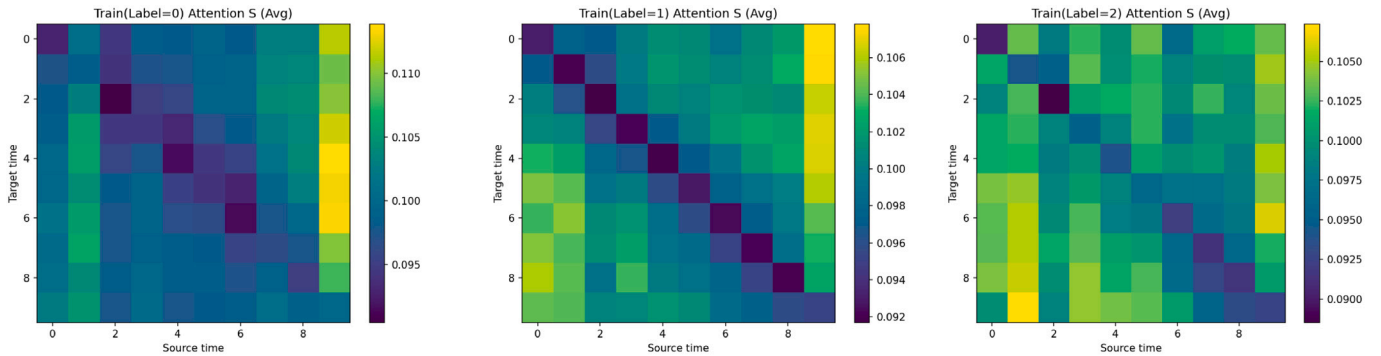
Behavior of non-sequential baselines and limitations of TabNet. For models that operate on flattened inputs (e.g., TabNet, SVM, RF, LGBM), the input tensor $X_i^{(t,\ell)} \in \mathbb{R}^{\ell \times 76}$ is reshaped into a vector of dimension 76ℓ . Tree-based methods (RF and LGBM) handle this representation relatively well but are ultimately surpassed by T-LSTM when the full 10-year history is utilized.

TabNet, despite being a state-of-the-art model for static tabular data, performs poorly in our experiments. A key reason is that TabNet is not explicitly designed to exploit temporal order. After flattening, the temporal structure is lost, and the model cannot naturally model dependencies along the time axis. Furthermore, the increase in dimensionality as ℓ grows (up to 760 features) makes optimization challenging. SVM shows a similar tendency, suffering from the curse of dimensionality. In contrast, T-LSTM explicitly encodes temporal structure and uses attention to highlight informative time steps, allowing it to fully leverage long sequences effectively.

4.2. Analysis of the importance of time steps in time-series features

To understand how T-LSTM utilizes temporal information and to enhance the interpretability of the model, we analyze the attention matrix $\bar{S}$. As described in Section 2, the model employs multi-head attention (with five heads); for interpretability, we compute the average of the attention matrices across all heads and use this averaged matrix in the following analysis. We focus on the 2019 experiment with $\ell = 10$.

Fig. 5 presents the heat map of the averaged attention matrix $\bar{S}$, defined from (2), for the training and test data. Each element $\bar{S}_{ij}$ represents the influence of time step j (source) on time step i (target). Indices 1 and 10 correspond to the most distant and the most recent years in the 10-year window, respectively.



(a) Label 0

(b) Label 1

(c) Label 2

Fig. 7. Heat map of $\bar{S}$ for each label of the training data (2019 Experiment).

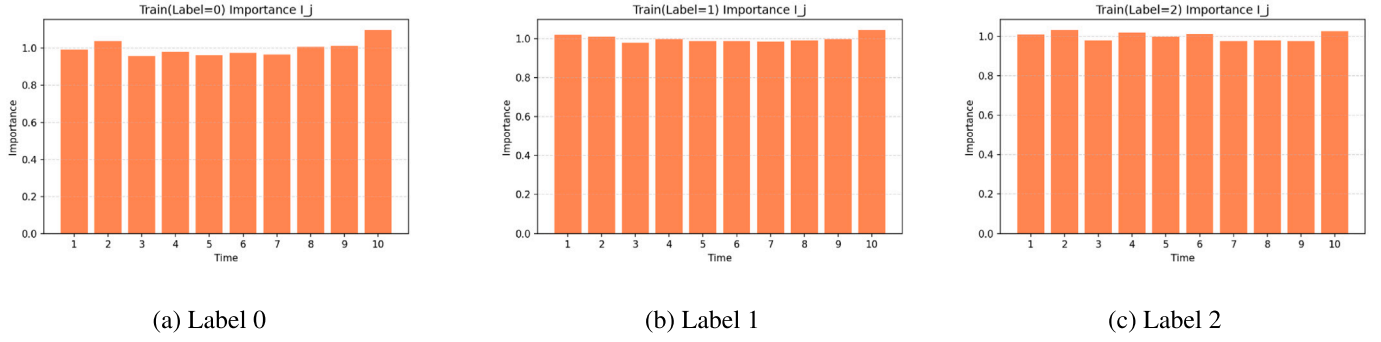


Fig. 8. Aggregated time step importance scores for the training data labels (2019 Experiment).

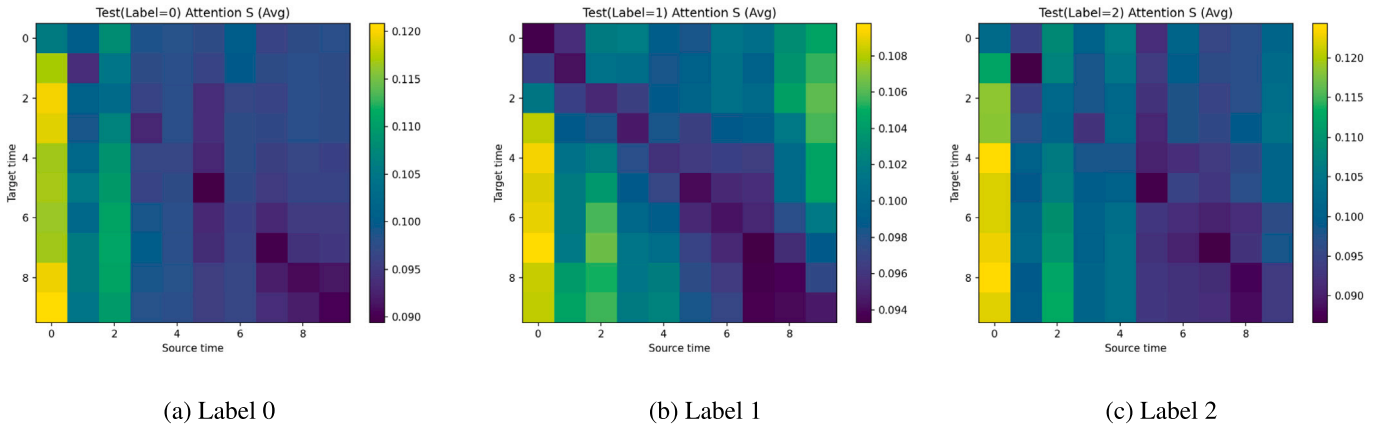


Fig. 9. Heat map of $\bar{S}$ for each label of the test data labels (2019 Experiment).

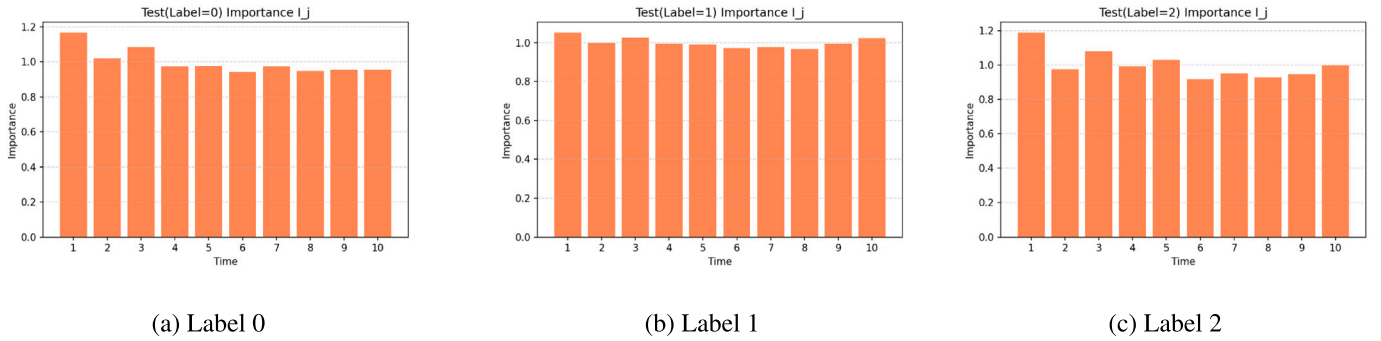


Fig. 10. Aggregated time step importance scores for test data labels (2019 Experiment).

To summarize the contribution of each year, we define the importance of time step j by

$$I_j = \sum_{i=1}^{10} \bar{S}_{ij}. \quad (13)$$

The aggregated importance values are shown in Fig. 6. We observe a pronounced U-shaped pattern: the first year ($j = 1$) and the most recent year ($j = 10$) have substantially higher importance than the intermediate years. This indicates that the model relies on both the long-term baseline financial condition at the beginning of the 10-year period and the current financial status when making credit rating decisions.

To further investigate how this temporal focus depends on the target class, we stratify the attention analysis by the class labels. Figs. 7–10 display the corresponding heat maps and importance scores I_j .

The visualization reveals distinct patterns depending on the credit rating class.

For Label 0 (highest credit quality), the model places high importance on both the earliest ($j = 1$) and the most recent ($j = 10$) years. This suggests that the model verifies long-term stability by confirming that both the historical baseline and the current status are strong.

For Label 2 (lowest credit quality), a similar emphasis on the first and last years is observed. This indicates that the model compares the long-term baseline with the current condition to identify firms with persistently weak or deteriorating performance over the decade.

In contrast, for Label 1 (intermediate credit quality), the importance scores are more concentrated on the most recent years ($j = 9$ and $j = 10$). This pattern suggests that the model relies more heavily on current financial information when firms lie between the highest and lowest rating classes, as their long-term evaluation may be less fixed.

These results show that the attention mechanism in T-LSTM not only improves predictive performance but also provides a transparent visualization of which specific years in the financial history are most influential for each rating class. This capability successfully identifies the temporal importance of input features and helps to alleviate the “black box” nature often associated with deep learning models, providing interpretable insights into the temporal aspects of corporate creditworthiness.

5. Conclusion

In this study, we propose the T-LSTM model to address the dual challenges of high classification accuracy and interpretability in corporate credit rating classification. Unlike traditional approaches that rely solely on current financial indicators, our method effectively integrates long-term longitudinal data to capture the dynamic trajectory of corporate financial health.

To rigorously evaluate the proposed model, we conducted comprehensive comparative experiments against state-of-the-art baselines, including LightGBM, TabNet, and Temporal Convolutional Networks (TCN), as well as standard machine learning algorithms. The experimental results demonstrated that the T-LSTM consistently outperforms these advanced methods. Notably, our analysis of varying time-series lengths revealed that the model achieves its most robust performance in the long-history setting (e.g., using a 10-year history of financial ratios), yielding an absolute accuracy improvement on the order of 1 percentage point when compared with a strong sequential baseline trained on the same 10-year history (e.g., LSTM) and up to about 17 percentage points when compared with representative single-year “snapshot” baselines using only the most recent year’s ratios. This finding underscores the critical importance of long-term dependencies in assessing corporate creditworthiness.

Furthermore, the T-LSTM offers significant advantages in terms of interpretability. By leveraging the attention mechanism, the model successfully visualizes the temporal importance of input features, identifying specific years that most significantly influence the credit rating

decision. This capability resolves the “black box” issue inherent in many deep learning architectures, providing financial analysts with transparent and actionable insights.

In conclusion, the T-LSTM serves as a powerful and reliable tool for the financial industry, enabling precise credit risk assessment while ensuring the accountability required for real-world applications. Future work will focus on extending this framework to include unstructured data, such as textual financial reports, to further enhance predictive performance.

CRedit authorship contribution statement

Mikio Kofune: Writing – review & editing, Formal analysis. **Kensei Monden:** Writing – original draft, Methodology, Investigation. **Suguru Yamanaka:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Definitions of financial indicators

Table A.5 lists the definitions and calculation formulas for all 76 financial indicators used in this study. In the formulas, terms such as “Net sales” and “Total assets” refer to the values obtained directly from the financial statements (income statement, balance sheet, and statement of cash flows). Ratios are expressed in level form (e.g., 0.25 corresponds to 25%), unless otherwise noted.

Table A.5
Definitions and formulas of the financial indicators.

Symbol	Indicator name	Definition/Formula
X_1	Net sales	Net sales as reported in the income statement.
X_2	Ordinary profit	Ordinary profit (operating profit plus non-operating income minus non-operating expenses) as reported in the income statement.
X_3	Total assets	Total assets as reported in the balance sheet.
X_4	Owners’ equity	Owners’ equity as reported in the balance sheet.
X_5	Cash flows from operating activities	Cash flows from operating activities as reported in the statement of cash flows (operating cash flows).
X_6	Net working capital	$X_6 = \text{Current assets} - \text{Current liabilities}$.
X_7	Current ratio	$X_7 = \frac{\text{Current assets}}{\text{Current liabilities}}$.
X_8	Asset ratio (quick ratio)	$X_8 = \frac{\text{Cash and deposits} + \text{Notes and accounts receivable}}{\text{Current liabilities}}$.
X_9	Current liabilities to cash flows from operating activities ratio	$X_9 = \frac{\text{Current liabilities}}{\text{Cash flows from operating activities}}$.
X_{10}	Fixed liabilities to cash flows from operating activities ratio	$X_{10} = \frac{\text{Fixed liabilities}}{\text{Cash flows from operating activities}}$.
X_{11}	Interest-bearing liabilities to cash flows from operating activities ratio	$X_{11} = \frac{\text{Interest-bearing liabilities}}{\text{Cash flows from operating activities}}$, where interest-bearing liabilities include short-term borrowings, bonds, long-term borrowings, and convertible bonds.
X_{12}	Debt-equity ratio	$X_{12} = \frac{\text{Total liabilities}}{\text{Owners’ equity}}$.
X_{13}	Equity ratio	$X_{13} = \frac{\text{Owners’ equity}}{\text{Total assets}}$.
X_{14}	Fixed assets to equity ratio	$X_{14} = \frac{\text{Fixed assets}}{\text{Owners’ equity}}$.

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Table A.5
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Symbol	Indicator name	Definition/Formula
X_{15}	Fixed assets to long-term capital ratio	$X_{15} = \frac{\text{Fixed assets}}{\text{Fixed liabilities} + \text{Owners' equity}}.$
X_{16}	Interest coverage ratio	$X_{16} = \frac{\text{Operating profit} + \text{Non-operating income}}{\text{Interest and discount expenses}}.$
X_{17}	Current liabilities to sales ratio	$X_{17} = \frac{\text{Net sales}}{\text{Current liabilities}}.$
X_{18}	Fixed liabilities to sales ratio	$X_{18} = \frac{\text{Net sales}}{\text{Fixed liabilities}}.$
X_{19}	Interest-bearing liabilities to sales ratio	$X_{19} = \frac{\text{Net sales}}{\text{Interest-bearing liabilities}}.$
X_{20}	Change ratio to sales ratio (sales growth rate)	$X_{20} = \frac{\text{Net sales}_t - \text{Net sales}_{t-1}}{ \text{Net sales}_{t-1} }.$
X_{21}	Change ratio of total profit (gross profit growth)	$X_{21} = \frac{\text{Gross profit}_t - \text{Gross profit}_{t-1}}{ \text{Gross profit}_{t-1} },$ where Gross profit = Net sales – Cost of goods sold.
X_{22}	Change ratio of operating profit	$X_{22} = \frac{\text{Operating profit}_t - \text{Operating profit}_{t-1}}{ \text{Operating profit}_{t-1} }.$
X_{23}	Change ratio of ordinary profit	$X_{23} = \frac{\text{Ordinary profit}_t - \text{Ordinary profit}_{t-1}}{ \text{Ordinary profit}_{t-1} }.$
X_{24}	Change ratio of total assets	$X_{24} = \frac{\text{Total assets}_t - \text{Total assets}_{t-1}}{\text{Total assets}_{t-1}}.$
X_{25}	Change ratio of owners' equity	$X_{25} = \frac{\text{Owners' equity}_t - \text{Owners' equity}_{t-1}}{ \text{Owners' equity}_{t-1} }.$
X_{26}	Gross profit margin	$X_{26} = \frac{\text{Gross profit}}{\text{Net sales}}.$
X_{27}	Operating profit to sales ratio	$X_{27} = \frac{\text{Operating profit}}{\text{Net sales}}.$
X_{28}	Ordinary profit to sales ratio	$X_{28} = \frac{\text{Ordinary profit}}{\text{Net sales}}.$
X_{29}	Pretax income to sales ratio	$X_{29} = \frac{\text{Pretax net earnings for the period}}{\text{Net sales}}.$
X_{30}	Administrative expense ratio	$X_{30} = \frac{\text{Selling, general and administrative expenses}}{\text{Net sales}}.$
X_{31}	Ordinary profit to total capital ratio	$X_{31} = \frac{\text{Ordinary profit}}{\text{Total capital}},$ where Total capital = Total liabilities + Owners' equity (equal to total assets).
X_{32}	Operating profit to total capital ratio	$X_{32} = \frac{\text{Operating profit}}{\text{Total capital}}.$
X_{33}	Business profit to total capital ratio	$X_{33} = \frac{\text{Operating profit} + \text{Non-operating income}}{\text{Total capital}}.$
X_{34}	Ordinary profit to owners' equity ratio	$X_{34} = \frac{\text{Ordinary profit}}{\text{Owners' equity}}.$
X_{35}	Operating profit to owners' equity ratio	$X_{35} = \frac{\text{Operating profit}}{\text{Owners' equity}}.$
X_{36}	Cash flows from operating activities to sales ratio	$X_{36} = \frac{\text{Cash flows from operating activities}}{\text{Net sales}}.$
X_{37}	Total assets turnover	$X_{37} = \frac{\text{Net sales}}{\text{Total assets}}.$
X_{38}	Equity turnover	$X_{38} = \frac{\text{Net sales}}{\text{Owners' equity}}.$
X_{39}	Trade receivables turnover	$X_{39} = \frac{\text{Net sales}}{\text{Notes and accounts receivable}}.$
X_{40}	Inventories turnover	$X_{40} = \frac{\text{Net sales}}{\text{Inventories}}.$
X_{41}	Current assets turnover	$X_{41} = \frac{\text{Net sales}}{\text{Current assets}}.$
X_{42}	Fixed assets turnover	$X_{42} = \frac{\text{Net sales}}{\text{Fixed assets}}.$
X_{43}	Change rate of liabilities	$X_{43} = \frac{\text{Total liabilities}_t - \text{Total liabilities}_{t-1}}{ \text{Total liabilities}_{t-1} }.$
X_{44}	Change rate of interest-bearing liabilities	$X_{44} = \frac{\text{Interest-bearing liabilities}_t - \text{Interest-bearing liabilities}_{t-1}}{ \text{Interest-bearing liabilities}_{t-1} }.$
X_{45}	Change rate of cash and deposits	$X_{45} = \frac{\text{Cash and deposits}_t - \text{Cash and deposits}_{t-1}}{ \text{Cash and deposits}_{t-1} }.$
X_{46}	Change rate of fixed assets	$X_{46} = \frac{\text{Fixed assets}_t - \text{Fixed assets}_{t-1}}{ \text{Fixed assets}_{t-1} }.$
X_{47}	Cash to deposits ratio	$X_{47} = \frac{\text{Cash and deposits}}{\text{Current liabilities}}.$
X_{48}	Discounted interest expense to sales ratio	$X_{48} = \frac{\text{Interest and discount expenses}}{\text{Net sales}}.$
X_{49}	Total profit to discounted interest expense ratio	$X_{49} = \frac{\text{Gross profit}}{\text{Interest and discount expenses}}.$

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Table A.5
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Symbol	Indicator name	Definition/Formula
X_{50}	Tangible fixed assets to total assets ratio	$X_{50} = \frac{\text{Tangible fixed assets}}{\text{Total assets}}$
X_{51}	Operating expenses to sales ratio	$X_{51} = \frac{\text{Operating expenses}}{\text{Net sales}}$, where operating expenses include cost of goods sold and selling, general and administrative expenses.
X_{52}	Other income to sales ratio	$X_{52} = \frac{\text{Non-operating income}}{\text{Net sales}}$
X_{53}	Other expenses to sales ratio	$X_{53} = \frac{\text{Non-operating expenses}}{\text{Net sales}}$
X_{54}	Interest burden to sales ratio	$X_{54} = \frac{\text{Interest and discount expenses} - \text{Interest and dividend income}}{\text{Net sales}}$, where net interest expenses are defined as interest and discount expenses minus interest and dividend income.
X_{55}	Ratio of current liabilities to cash flows from operating activities	$X_{55} = \frac{\text{Current liabilities}}{\text{Cash flows from operating activities}}$
X_{56}	Ratio of current liabilities to owners' equity (owners' equity/operating CF)	$X_{56} = \frac{\text{Current liabilities}}{\text{Owners' equity}}$
X_{57}	Total capital growth ratio	$X_{57} = \frac{\text{Total capital}_t - \text{Total capital}_{t-1}}{ \text{Total capital}_{t-1} }$
X_{58}	Pretax net earnings for the period to total capital ratio	$X_{58} = \frac{\text{Pretax net earnings for the period}}{\text{Total capital}}$
X_{59}	Tangible fixed assets growth ratio	$X_{59} = \frac{\text{Tangible fixed assets}_t - \text{Tangible fixed assets}_{t-1}}{ \text{Tangible fixed assets}_{t-1} }$
X_{60}	Liabilities turnover period	$X_{60} = \frac{\text{Total liabilities}}{\text{Net sales}}$
X_{61}	Current liabilities turnover period	$X_{61} = \frac{\text{Current liabilities}}{\text{Net sales}}$
X_{62}	Short-term borrowings turnover period	$X_{62} = \frac{\text{Short-term borrowings}}{\text{Net sales}}$
X_{63}	Capital turnover period	$X_{63} = \frac{\text{Total capital}}{\text{Net sales}}$
X_{64}	Fixed liabilities turnover period	$X_{64} = \frac{\text{Fixed liabilities}}{\text{Net sales}}$
X_{65}	Days cash and deposits on hand	$X_{65} = \frac{\text{Cash and deposits}}{\text{Net sales}}$, which can be interpreted as the number of years of net sales covered by cash and deposits.
X_{66}	Cash interval	$X_{66} = \frac{\text{Cash and deposits}}{\text{Selling, general and administrative expenses}}$
X_{67}	Current assets to total assets ratio	$X_{67} = \frac{\text{Current assets}}{\text{Total assets}}$
X_{68}	Fixed assets to total assets ratio	$X_{68} = \frac{\text{Fixed assets}}{\text{Total assets}}$
X_{69}	Inventories to total assets ratio	$X_{69} = \frac{\text{Inventories}}{\text{Total assets}}$
X_{70}	Pretax net earnings for the period to total assets ratio	$X_{70} = \frac{\text{Pretax net earnings for the period}}{\text{Total assets}}$
X_{71}	Logarithm of total assets	$X_{71} = \ln(\text{Total assets})$
X_{72}	Tangible fixed assets ratio	$X_{72} = \frac{\text{Tangible fixed assets}}{\text{Owners' equity}}$
X_{73}	Legal capital to owners' equity ratio	$X_{73} = \frac{\text{Legal capital}}{\text{Owners' equity}}$, where legal capital corresponds to paid-in capital (capital stock).
X_{74}	Financial leverage ratio	$X_{74} = \frac{\text{Total assets}}{\text{Owners' equity}}$
X_{75}	Tangible fixed assets turnover	$X_{75} = \frac{\text{Net sales}}{\text{Tangible fixed assets}}$
X_{76}	Net income growth ratio (pretax)	$X_{76} = \frac{\text{PNE}_t - \text{PNE}_{t-1}}{ \text{PNE}_{t-1} }$, where PNE _{<i>t</i>} denotes pretax net earnings for the period in year <i>t</i> .

Data availability

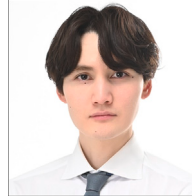
The authors do not have permission to share data.

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